Subword RNR as Multidominance-Plus-Ellipsis

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Abstract: This paper investigates subword Right-Node Raising (RNR), a variant of RNR where the element shared between conjuncts is smaller than a word. As for phrasal RNR, diagnostics show that subword RNR can correspond to two different underlying structures, multidominance and ellipsis. Based on the data from subword RNR, we subject the ellipsis approach to RNR to close scrutiny. We show that existing analyses of RNR-as-ellipsis fail to extend to subword RNR, and we propose a modified analysis according to which a focus-marked element in backward ellipsis can survive ellipsis in situ. Our proposal captures both phrasal and subword RNR in a unified framework.
Keywords: *Right-Node Raising*, *multidominance*, *ellipsis*, *in-situ survival*, *focus*

1 Introduction

Right-Node Raising (RNR) is a construction in which an element, typically realized at the right edge of the coordination, is semantically interpreted as part of both conjuncts (1).

(1) Yin likes, but Yosho hates, pig trotters.

We will refer to the shared element (here underlined) as the pivot, and to the first and second conjunct at the exclusion of the pivot as remnant and correlate, as schematized in (2):

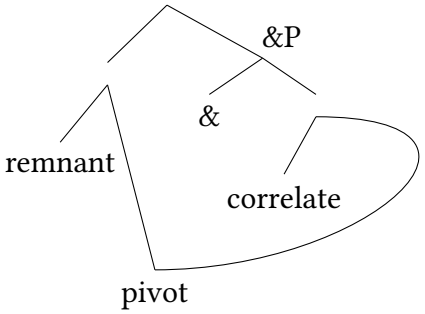
(2) remnant- and correlate-pivot

Previous work has argued that RNR can realize two different underlying structures: either a multidominance structure (RNR-MD) or an ellipsis structure (RNR-E) (Abels 2004, Barros &

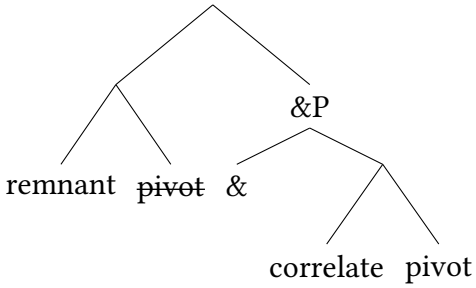
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Vicente 2011, Belk, Neeleman & Philip 2023, Citko 2017, a.o.). In RNR-MD, the pivot is multidominated across conjuncts and spelled out only once at the right edge of the coordination due to a linearization algorithm (3). By contrast, in RNR-E, the pivot is underlyingly present in all conjuncts but is elided or deleted in all but the final conjunct (4).

(3)



(4)



Against this background, the present paper is concerned with what we will refer to as subword RNR, i.e., instances of RNR such as (5) in which the pivot is smaller than a word:

(5) over- and underestimate

Using the same diagnostics that have been employed to support a dual analysis of phrasal RNR, we begin by showing that subword RNR equally needs to receive a dual analysis as either RNR-MD or RNR-E. In the remainder of this paper, we then focus on RNR-E. Previous ellipsis analyses of RNR have often left the nature of the deletion operation involved unspecified. We contrast two different deletion analyses: prosodic deletion – a purely morphophonological deletion operation targeting a string or a prosodic constituent – and syntactically licensed ellipsis. We show that certain cases of subword RNR cannot be derived through prosodic deletion, but that existing approaches to syntactically licensed ellipsis also fail to account for these data. In response, we develop a novel approach to RNR-E, focusing mainly on the verbal domain. On our analysis, ellipsis is syntactically licensed and targets the *v*P, as in ordinary VP ellipsis (Aelbrecht 2010, Merchant 2001, a.o.). What is new is that focus-marked verbal heads, and their subword parts, can survive ellipsis in situ. This captures not only our subword data but also phrasal RNR data that previous accounts struggle to explain.

Our new proposal for ellipsis is specifically designed for backward ellipsis (RNR-E). We argue that by contrast, forward ellipsis, such as VP ellipsis, does not allow for in-situ survival; instead, remnants must move out of the ellipsis site to escape elision. This contrast correctly derives certain asymmetries between forward and backward ellipsis. We end this paper by discussing why forward and backward ellipsis might differ in the mechanisms for remnant survival, touching on prosodic realization and sentence processing.

We proceed as follows. Section 2 introduces the dual analysis of RNR as either RNR-MD or RNR-E, also extending the diagnostics originally introduced for phrasal RNR to subword RNR. Zooming in on RNR-E, we then critically review previous ellipsis analyses of RNR (Section 3), before presenting our own proposal (Section 4). Section 5 discusses differences between backward ellipsis (RNR-E) and forward ellipsis (VP ellipsis), and Section 6 concludes.

2 The dual analysis of RNR

We begin by reviewing the diagnostics that Barros & Vicente (2011) and Belk, Neeleman & Philip (2023), a.o., have taken to be indicative of RNR-MD or RNR-E. Evidence for RNR-MD comes from cases of RNR that are either morphologically or semantically different from the corresponding full coordination and thus cannot be derived from the latter via ellipsis or deletion. First, RNR allows for cumulative agreement as in (6a), where the verb surfaces with plural marking despite the fact that in the corresponding full coordination, the verbs in the individual conjuncts are singular (6b). Thus, (6a) cannot be derived from (6b) by deleting part of the first conjunct.

(6) a. Sue's proud that Bill_{SG}, and Mary's glad that John_{SG}, have $_{\text{PL}}$traveled to Cameroon.
b. Sue's proud that Bill_{SG} has_{SG} traveled to Cameroon, and Mary's glad that John_{SG} has_{SG} traveled to Cameroon. (Grosz 2015:6)

In other cases, an ellipsis analysis is ruled out because the RNR example allows for a semantic interpretation that is not available in the full coordination. For instance, (7a) but not (7b) allows for an internal reading of *same* according to which the tune whistled by John is identical to the one hummed by Mary:

(7) a. John whistled, and Mary hummed, the same tune. (Jackendoff 1972:192)
b. John whistled the same tune, and Mary hummed the same tune.

Similarly, the RNR example (8a) is compatible with a distributive plural reading according to which Ava and Habib only study one Oceanic language each, which is not possible in (8b):

(8) a. Ava will soon, and Habib is going to, study Oceanic languages.
b. Ava will soon study Oceanic languages, and Habib is going to study Oceanic languages. (Belk, Neeleman & Philip 2023:700)

Distributive plural interpretations have also been observed for subword RNR (Artstein 2005). In (9a), it is possible that William is only a sociolinguist and Lila only a psycholinguist; again, the relevant reading is unavailable in the corresponding full coordination (9b):

(9) a. William and Lila are socio- and psycholinguists.
b. William and Lila are sociolinguists and psycholinguists.

To summarize, the RNR-MD analysis has been proposed for instances of RNR that cannot be derived from the corresponding full coordination because the two are either morphologically or semantically distinct (Barros & Vicente 2011, Belk, Neeleman & Philip 2023, see also Abels 2004, Grosz 2015, Sabbagh 2007, a.o.). Since our focus in this paper is on RNR-E, we leave aside the details of the multidominance analysis and whether it successfully captures the data seen so far; for discussion, see Neu & Guo (2026).

The RNR-E analysis has been proposed for instances of RNR where the missing material in the first conjunct is not morphophonologically identical to the overt pivot. In (10), the pivot contains the form *waking*, but the verb in the first conjunct would surface as *wake*:

(10) I usually fail to ~~wake up early~~, but Ava always succeeds in waking up early.

(Belk, Neeleman & Philip 2023:689f.)

Similar evidence comes from vehicle change. In (11), the pivot contains the R-expression *Ava*, but the corresponding form in the first conjunct would need to surface as the pronoun *her* to avoid a Principle C violation:

(11) She_1 hopes that he won't ~~fire her~~$_1$, but I fear that the boss will fire Ava_1.

(Belk, Neeleman & Philip 2023:689f.)

Parallel examples of morphological mismatches and vehicle change effects can be found for subword RNR (12):

(12) a. Yosho has over-~~estimated the cost of the plan~~ and Yin will underestimate the cost of the plan.
b. She_1 hopes that he will over~~pay her~~$_1$, but I know that the boss will underpay Ava_1.

Under RNR-MD, these data are unexpected: if the pivot is present only once in the derivation, it should not differ in the two conjuncts. By contrast, it is well-known that ellipsis tolerates morphological mismatches and vehicle change, shown for VP ellipsis in (13):

(13) a. Ava always succeeds in waking up early, but I usually fail to ~~wake up early~~.
b. I fear that the boss will fire Ava, although she hopes that he won't ~~fire her~~.

(Belk, Neeleman & Philip 2023:689f.)

Accordingly, morphological mismatches and vehicle change in RNR have been taken as evidence for RNR-E (Barros & Vicente 2011, Belk, Neeleman & Philip 2023, Bošković 2004, Citko 2017, Ha 2008, a.o.).

While the general motivation behind RNR-E is clear, the question remains what exactly the nature of the ellipsis operation in question is. In the next section, we review some previous answers to this question that have been given in the literature, showing that they fall short of accounting for the full set of data. In Section 4, we then propose our alternative analysis.

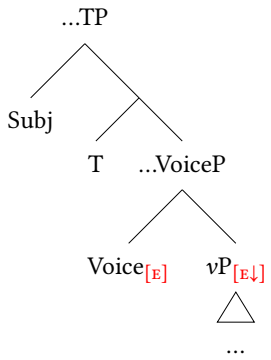
3 Previous approaches to RNR-E

The previous literature on RNR has used "ellipsis" as a largely descriptive term, without providing a precise characterization of which elements are targeted by deletion and how the deletion is licensed. Barros & Vicente (2011: 1), for instance, "take [RNR-E] to involve PF deletion of a syntactically and semantically complete VP, rather than insertion of a verbal proform." Belk, Neeleman & Philip (2023: 715) treat ellipsis as an instance of nonrealization at PF, and require its licensing to satisfy three conditions: (i) directionality, i.e., the elided material precedes its postcedent; (ii) parallelism, i.e., the elided material and its postcedent are focus alternatives to each other (following Hartmann 2001); and (iii) identical ordering, i.e., the elided material satisfies the same ordering statement as the pivot. Although Barros &

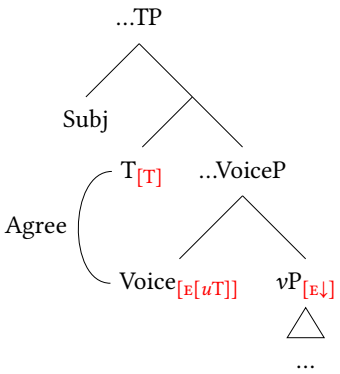
Vicente (2011) and Belk, Neeleman & Philip (2023) propose different conceptions of RNR-E, these do not necessarily differ in empirical coverage. The key issue is how data showing properties of either *prosodic deletion* or *syntactically licensed ellipsis* can be subsumed under the descriptive label "ellipsis" both approaches use.

Under the prosodic deletion approach, PF-deletion is string-based: it applies to any pair of peripheral strings that share the same morph forms and are prosodically independent (Chaves 2014:838; see also Hartmann 2001, Wilder 1997, a.o.). Under the ellipsis approach, elision is sensitive to syntactic structure and typically targets a syntactic constituent. Following Zagona (1988) and Lobeck (1995, 1999), later adopted by Merchant (2001) and others, ellipsis is licensed when the licensing head and the ellipsis site stand in a head–complement configuration, as in (14). Aelbrecht (2010), by contrast, proposes that the licensor enters into an Agree relation with an intermediate functional head bearing an [E]-feature, which in turn licenses the deletion of its complement, as in (15). In what follows, a constituent bearing [E↓] is understood to be unpronounced at PF: under both (14) and (15), *v*P is elided.

(14)



(15)



In the following, we use the term "ellipsis" specifically for syntactically licensed ellipsis.

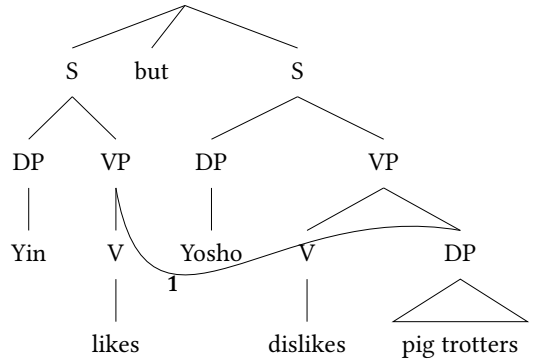
For the subword RNR data examined here, neither prosodic deletion nor ellipsis, as currently formulated, captures the facts adequately. First, Wilder (1997) argues that prosodic deletion is assumed to require strict morphophonological identity between the deleted material and its postcedent. Accordingly, it cannot capture data involving morphological mismatches and vehicle change, as seen in (12).

At the same time, ellipsis fails as well. Syntactically licensed ellipsis is expected to target syntactic constituents; specifically, *v*P. In (12), the string in the first conjunct corresponding to the pivot – *-estimated the cost of the plan* and *-pay her* – does not form a syntactic constituent, because it combines a subword part with a DP. It could form a constituent only if the subword were evacuated from the *v*P, leaving a trace inside the ellipsis site, so that what looks like a non-constituent on the surface is in fact a constituent (in the same vein as Kayne 1995). However, extracting a subword out of a word is generally banned, so there is no way to move the subword out of the ellipsis site and make the elided string in (12) a syntactic constituent.

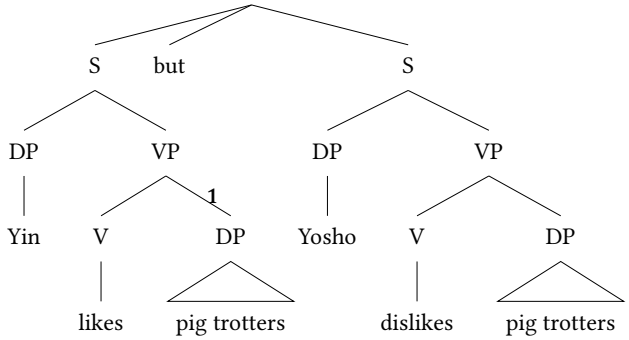
In addition to prosodic deletion and ellipsis, Belk, Neeleman & Philip (2023) propose a pruning operation, which serves as a descriptive device for what happens at PF. Specifically, they derive both multidominance patterns and ellipsis patterns (in the descriptive sense) with this single operation. Pruning obligatorily removes a branch that crosses other branches (mul-

tidominance), such as branch 1 in (16), and optionally removes a branch that is syntactically and semantically recoverable (ellipsis), such as branch 1 in (17).

(16) Pruning for MD



(17) Pruning for ellipsis



As currently formulated, however, pruning offers no advantage over ellipsis on the two points that concern us here. First, in the pruning exemplifications discussed by Belk, Neeleman & Philip (2023), the pruned branch is phonologically identical to its postcedent. Yet RNR also permits morphological mismatch between the two conjuncts. To extend pruning to such cases, one would have to claim that a surface morphological mismatch nonetheless reflects syntactic identity, which is exactly what syntactically licensed ellipsis assumes (Lasnik 1995, Merchant 2007, a.o.). Pruning therefore gains no advantage over ellipsis on this point. Second, although pruning of branch 1 in (17) is empirically well motivated, it remains unclear why the same operation is unavailable in the forward counterpart, **Yin likes pig trotters, but Yosho dislikes*. Without an account of this asymmetry, pruning has no theoretical edge over ellipsis, which faces the same directionality restriction.

We therefore adopt ellipsis for RNR. A PF-deletion operation such as pruning may eventually offer a better analysis, but as currently formulated it faces the same limitations as ellipsis, which is far better explored. Although ellipsis, like prosodic deletion and pruning, does not fully capture the subword RNR data in (12), we propose a modified version of ellipsis that retains its core insight: syntactically licensed ellipsis tolerates morphological mismatch and vehicle change.

4 Proposal: focus-marked material can survive backward ellipsis in situ

In this section, we argue that focus-marked elements can survive ellipsis in situ in RNR. That focused elements survive ellipsis in some form is widely accepted. Under the account in Takahashi & Fox (2005) (see also Griffiths 2019, Hartmann 2001, Rooth 1992a,b, a.o.), ellipsis is subject to a parallelism condition (19); that is, an identity requirement that applies within a parallelism domain (18):

(18) For ellipsis of EC [elided constituent] to be licensed, there must exist a constituent, which reflexively dominates EC, and satisfies the parallelism condition in (19). [Call this constituent the parallelism domain (PD).] (Takahashi & Fox 2005:229)

(19) Parallelism
PD satisfies the parallelism condition if PD is semantically identical to another constituent AC, modulo focus-marked constituents. (Takahashi & Fox 2005:229)

Crucially, focus-marked material is excluded when parallelism is computed. Ellipsis is therefore licensed whenever the non-focus-marked material in the ellipsis site and its antecedent/postcedent satisfies the parallelism condition (19), and focus-marked elements do not prevent it from applying. Focus-marked material is accordingly stranded when ellipsis applies.

How focused material survives ellipsis is debated. Two different approaches have been developed in the literature. On the first, remnants survive by moving out of the ellipsis site before ellipsis applies (ex-situ survival; see Merchant 2001, 2004, Sailor & Thoms 2014, Weir 2014, Yoshida, Nakao & Ortega-Santos 2015, a.o.), as in (20a). On the second, the surviving material remains in situ and ellipsis targets the material around it (in-situ survival; see Bruening 2015, Griffiths 2019, Griffiths, Güneş & Lipták 2023, Ott & Struckmeier 2018, a.o.), as in (20b).

(20) a. Remnant$_i$ ~~[X t$_i$ Y]~~$_{\text{ellipsis site}}$
b. [~~X~~ Remnant ~~Y~~]$_{\text{ellipsis site}}$

In support of the ex-situ approach, Thoms (2010) has argued that a verb's ability to survive ellipsis is tied to its ability to raise above the ellipsis site. English auxiliary verbs can raise above *v* and therefore survive VP ellipsis, whereas main verbs cannot raise above *v* (Emonds 1978, Pollock 1989) and therefore cannot be stranded. The contrast between (21a) and (22a) shows that only auxiliaries can move to T, and this ability mirrors head stranding patterns: auxiliaries can be stranded on the surface (22b), whereas main verbs cannot (21b).

(21) a. *Put Morag a book on the table? *Yes-no questions*
b. *Morag put a book on the table, and Rab put, too.
c. Morag put a book on the table, and Rab did, too. (Thoms 2010)

(22) a. Is Morag a fool?
b. Morag is a fool, and Rab is, too. (Thoms 2010)

We adopt Thoms's (2010) proposal that under VP ellipsis in the second conjunct of a coordination, as in (21c) and (22b), henceforth *forward ellipsis* (as opposed to *backward ellipsis*, where ellipsis occurs in the first conjunct, as in RNR), a verbal head can survive only if it can be extracted from the ellipsis site, that is, escape ellipsis ex situ. In what follows, we will argue that verbal heads in RNR can instead survive in situ, which complements the literature on head stranding and ellipsis.

Returning to our data in (12a) involving backward ellipsis, repeated below as (23), a subword cannot undergo movement, so ex-situ survival is ruled out for deriving (23) under an ellipsis approach. We thus argue that the focus-marked subword *over-* survives ellipsis in situ.

(23) Yosho has over-~~estimated the cost of the plan~~ and Yin will underestimate the cost of the plan.

In-situ survival applies not only to subword RNR but also to RNR with a stranded full verb, as in (24a). Most previous literature assumes that cases like (24a) can only be captured by RNR-MD

since they have no forward-ellipsis counterpart (24b): as seen above, main verbs cannot move out of the ellipsis site under forward ellipsis and hence cannot be stranded.

(24) a. Yosho will overestimate ~~the cost of the plan~~ and Yin will underestimate the cost of the plan.
b. *Yosho will overestimate the cost of the plan and Yin will underestimate ~~the cost of the plan~~.

Cases like (25a) and (26a), however, are problematic for both multidominance and traditional ellipsis. They are assumed to be beyond the reach of multidominance, since multidominance involves a single token of a shared constituent across both conjuncts, which should not allow different phonological forms. They are also beyond the reach of traditional ellipsis, since their forward-ellipsis counterparts (25b) and (26b) are ungrammatical: verbs cannot be stranded outside the VP ellipsis site in forward ellipsis. Yet (25a) and (26a) show characteristics of forward VP ellipsis, tolerating both vehicle change (25c) and polarity-sensitive indefinite DP mismatch (26c).

(25) a. She$_1$ hopes that he will overpay ~~her~~$_1$, but I know that the boss will underpay Ava$_1$.
b. *She$_1$ hopes that he will overpay her$_1$, but I know that the boss will underpay ~~Ava~~$_1$.
c. She$_1$ hopes that he will overpay her$_1$, but I know that the boss will not ~~overpay Ava~~$_1$.

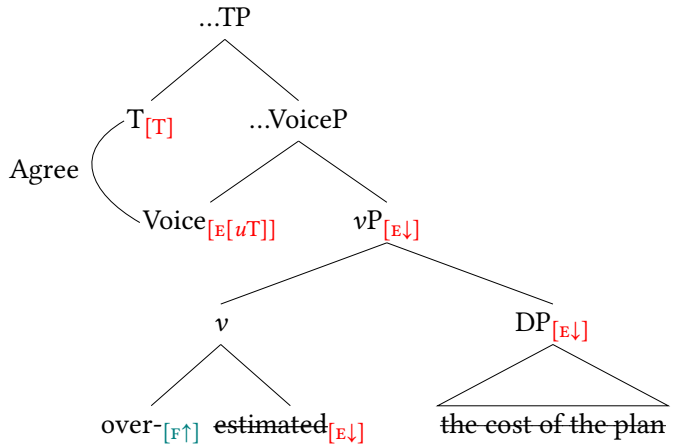
(26) a. John devoured ~~some of the cakes~~, but Mary didn't devour any of the cakes.
b. *John devoured some of the cakes, but Mary didn't devour ~~any of the cakes~~.
c. John devoured some of the cakes, but Mary didn't ~~devour any of the cakes~~.

Our modified version of ellipsis, in which focus-marked material survives ellipsis in situ, captures not only subword cases like (23) but also RNR cases with stranded material above the word level, such as (25a) and (26a). In (25a), *overpay* is focus-marked in contrast to *underpay*, and in (26a), *devour* is focus-marked in contrast to *didn't devour*. Being focus-marked, these verbs escape muting and survive in situ. In their forward-ellipsis counterparts, by contrast, a remnant can survive ellipsis only by extracting from the ellipsis site, which main verbs cannot do. This rules out (25b) and (26b). Our analysis thus offers a unified account of subword and phrasal RNR data that show characteristics of ellipsis but have no forward-ellipsis counterpart, data that, in the previous literature, neither multidominance nor traditional ellipsis adequately captures.

To see how our analysis works in detail, we assume that ellipsis in (23) is licensed by the same system as canonical VP ellipsis in the second conjunct of a coordination, since the two show similar patterns (see Bošković 2004, a.o.). Following Aelbrecht (2010), we take T to be the licensor for VP ellipsis, with a long-distance Agree relation between T and Voice: the [E]-feature on Voice is activated once its uninterpretable [T]-feature is checked. Activation of [E]-feature on Voice sends its complement, *v*P, to PF, where it is muted (marked as [E↓]), except for focus-marked elements (marked as [F↑]). We assume that an [F↑]-bearing element

shields itself from [E↓]-induced muting, in the spirit of Landau (2020). The same structure applies to (25a) and (26a), except that the whole verb bears [F↑] and therefore escapes muting.

(27)



Contrast in polarity, main verb or modality, tense, and aspect makes these elements candidates for bearing focus, and hence for surviving ellipsis in situ in RNR. However, contrastive focus is not the only type of focus that qualifies an element for in-situ survival (see also Chaves 2014). We find that additive focus (König 1977, 1981, Zimmermann 2014, a.o.) also does so, as the following examples show. Judgments on (28a) may vary, but the addition of *also* clearly improves it relative to (28b), which is completely ruled out.

(28) a. %Yosho has over-~~estimated the cost of the plan~~, and Yin will **also** over-estimate the cost of the plan.

b. *Yosho has over-~~estimated the cost of the plan~~, and Yin will over-estimate the cost of the plan.

We assume that (28a) is judged better than (28b) because *also* triggers focus alternatives, which allow a pitch accent to be realized in the second conjunct of (28a), where the material would otherwise be given and deaccented, as in (28b). We leave the precise mechanism for future research.

Note that, although this study focuses mainly on verbal material at both the subword and phrasal levels, as in (23), (24a), (25a) and (26a), in-situ survival is not limited to verbs or their subparts. Adverbial material can also be stranded, although such stranding is subject to a property specific to RNR, the Right Edge Condition (Sabbagh 2007, Wilder 1997, 2008, a.o.), as shown in (29). This condition requires the gap in a non-final conjunct, such as A in (29), to occur at the right edge of its conjunct.

(29) [[$_A$...gap] Conj [$_B$...pivot]]

Specifically, the ungrammaticality of (30a) is not a counterargument to in-situ survival, since the independent requirement in (29) rules it out. As (30b) shows, when a focus-bearing adverbial is stranded to the left of the gap rather than to the right, it can survive ellipsis in situ, unless one assumes that such adverbials are base-generated outside the ellipsis site, *v*P.

(30) a. *Yosho has OVER-~~estimated the cost of the plan~~ GREATLY, and Yin will UNDER-estimate the cost of the plan SLIGHTLY.
b. Yosho has GREATLY OVER-~~estimated the cost of the plan~~, and Yin will SLIGHTLY UNDER-estimate the cost of the plan.

This raises a further question: why the edge effect should hold for backward but not for forward ellipsis. Compare (31) with (30a). Adverbials can be stranded at the right edge of the second conjunct (31), but not at the right edge of the first conjunct in an RNR with a gap preceding it (30a).

(31) Yosho has overestimated the cost of the plan GREATLY, and Yin will ~~overestimate the cost of the plan~~ SLIGHTLY.

We do not have a definitive answer as to why backward ellipsis, unlike forward ellipsis, is sensitive to an edge effect. This effect has been taken as a chief advantage of the LCA-based multidominance approach to RNR (Wilder 1999, 2008), which derives the Right Edge Condition and directionality (the gap precedes the pivot) in an RNR. But ellipsis involves no LCA-driven linearization of shared material, yet RNR-E shows the same edge effect as RNR-MD. The effect therefore need not be derived by LCA-multidominance. It may instead reflect a deeper structural asymmetry between the first and second conjuncts of a coordination, which would weaken a key argument in favour of the LCA-multidominance approach.

In short, this section argues for a modified ellipsis approach to RNR under which elements bearing contrastive or additive focus survive ellipsis in situ, unlike in forward ellipsis, where remnants survive ex situ. We draw mainly on data involving subparts of verbs and full verbs and focus on the verbal domain, but the analysis is not limited to these cases: other elements, such as adverbials, can also survive ellipsis in situ in RNR, provided that the special properties of RNR, such as the Right Edge Condition, are respected. The novel analysis unifies subword and phrasal RNR under a single operation, syntactically licensed ellipsis, and captures data that previous multidominance-plus-(traditional-)ellipsis approaches cannot.

5 Towards deriving differences between forward and backward ellipsis

The proposal presented so far raises the question why the directionality of ellipsis should affect whether the remnant can survive ellipsis in situ. While solving this question conclusively is beyond the scope of this paper, we suggest that this asymmetry might reflect an interaction between directionality and sentence processing.

In RNR-E, the postcedent follows the ellipsis site, so the parser must maintain an unresolved dependency until it is encountered. We assume that this poses a challenge during sentence processing. RNR is typically accompanied by pitch accent on the stranded material and its correlate and by a rising boundary tone at the end of each conjunct. Together, these cues may ease the processing difficulty by signalling that the structure is incomplete and allowing the parser to anticipate the postcedent (Kentner, Féry & Alter 2008, Ott 2025, Selkirk 2002, a.o.). But note also that a rising boundary tone alone does not make (32a) acceptable, in contrast to its forward counterpart (32c), which is fully acceptable without any such tone. RNR becomes possible once a corresponding *do* is added in the second conjunct (32b). We assume that

does in (32b) provides a host for the pitch accent, allowing it to be realized in parallel on the stranded element in the first conjunct, *don't*, and on its correlate in the second, *does*.

(32) a. ??I don't ~~like sweet things~~, but Mary likes sweet things.
b. I don't ~~like sweet things~~, but Mary does, <u>like sweet things</u>.
c. Mary likes sweet things, but I don't ~~like sweet things~~.

In other words, pitch accent and a rising boundary tone jointly ease the processing of backward dependencies in RNR-E, as many have argued (Chaves 2014, Kentner, Féry & Alter 2008, Ott 2025, Selkirk 2002, a.o.). What is new in our proposal is that pitch accent can be realized properly only on parallel elements: *don't* and *does* are parallel, whereas *don't* and *like* are not. The following examples also illustrate this. Stranding a *do*-element at the right edge is degraded in the first conjunct (33a) but not in the second (33b). When a lexical verb replaces the *do*-element, judgments improve relative to (33a), as in (33c). We take the contrast between (33a) and (33c) to reflect how pitch accent is realized, which depends on parallelism.

(33) a. ??Yin did ~~eat a pig trotter~~, and Yosho also ate a pig trotter.
b. Yin ate a pig trotter, and Yosho also did ~~eat a pig trotter~~.
c. Yin ate ~~a pig trotter~~, and Yosho also ate, <u>a pig trotter</u>.

We hypothesize that pitch accent in RNR can be carried only by parallel elements, although what exactly counts as parallel remains to be determined. The special prosodic requirements of RNR may explain why in-situ survival is possible. Pitch accent requires a host, hosts must be parallel, and accented elements are typically focus-marked. Focus-marked elements can escape ellipsis, but verbs and subwords cannot be extracted from the ellipsis site even when they bear focus, so they have to survive in situ. In forward ellipsis, by contrast, no such processing cost arises, no prosodic prominence cues are required, and in-situ survival is not needed.

A further puzzle is why backward ellipsis in subordination (34a) behaves exactly like its forward counterpart (34b), whereas backward RNR-E differs systematically from its forward counterparts.

(34) a. Before Yin did ~~reveal Eileen's crime to the public~~, Yosho revealed Eileen's crime to the public.
b. Before Yosho revealed Eileen's crime to the public, Yin did ~~reveal Eileen's crime to the public~~.

This is a difficult question, and we know of no direct solution in the literature. One could argue that the syntactic asymmetry between subordination and coordination gives rise to this contrast. We do not rule out this direction, but here we pursue a processing-based alternative. Backward dependencies are assumed to incur processing costs, yet such constructions can nonetheless be strikingly acceptable. We suggest that they differ in the cues used to ease those costs. RNR-E relies on pitch accent on parallel elements and a rising boundary tone, and in-situ survival may be correlated with this prosodic strategy. In subordination, by contrast, the presence of *before*, *after*, *when* and similar subordinators already signals that further material will follow, so no comparable prosodic prominence is needed, and the correlated

in-situ survival is not relevant. As in forward ellipsis, ex-situ survival is then the only option. We leave the precise nature of the cues in subordination for future investigation.

In sum, we suggest that the difference between forward and backward ellipsis in how elements survive may be attributed to the processing costs of backward dependencies and the cues that overcome them, neither of which arises in the same way in forward ellipsis. This may explain why in-situ survival is possible in RNR but not in its forward counterparts. The special prosodic prominence of RNR may also distinguish it from other backward-dependency constructions, which use different cues and therefore lack the correlated in-situ survival. Our understanding of the interaction between directionality and processing in RNR, its forward counterparts, and the related subordination cases is still at a very early stage, and more theoretical, experimental and cross-linguistic work is needed.

6 Conclusion

In this paper, we have used data from subword RNR to critically examine previous formalizations of the dual multidominance-plus-ellipsis approach to RNR. Focusing on ellipsis, we have shown that existing analyses fail to account for subword RNR. To resolve this problem, we have argued for a modified ellipsis approach under which focus-marked elements can survive ellipsis in situ, in contrast to the ex-situ survival found in forward ellipsis. This analysis captures the data from subword RNR as well as certain cases of phrasal RNR that pose a challenge for previous work. We have further suggested, albeit tentatively, that the asymmetry in element survival across directionality may be explained by processing and prosody, a direction that calls for further research.

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